

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing
Project Scribe Submission

Group No.: S1 G17

Domain: ITS

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Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Answer: We simulate a road cross with traffic controlled lights where arrivals of the vehicles are randomly distributed and hence the queue length and congestion as time varies. The arrival times of vehicles in a small time interval is approximated with the help of the Poisson random variable.

$$N_k \sim \text{Poisson}(\lambda Dt)$$

λ is the mean rate of arrivals.

The unknown nature of arrivals makes the queue length a random process. The aim of the system is to reduce the predicted length of queues $E[Q_k]$, because this is a direct measure of congestion and delay.

The uncertainty process is mainly the arrival process N_k . This uncertainty spreads to queue to the next time steps and subsequently to the state of congestion at the next time steps. In Milestone-2 we transitioned out of conceptual description of this behaviour to a mathematical form of the same.

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Answer: The following random variables define the unpredictability of the flow of traffic in the system:

n - Count of arrivals: the number of vehicles entering the intersection within a time period Dt which is a Poisson random variable with rate λ .

λ - Arrival rate: is the average traffic rate which may change over time.

$q_{in}(k)$ - Input flow: A volume of vehicles entering the system performed at time step k ; this is random due to the fact that arrivals are random.

$q_{out}(k)$ -Output flow: number of vehicles out of the system at time step k , a result of current queue plus signal decision $u(k)$.

$Q(k)$ - Queue length: system state which varies overtime according to arrivals and departures.

Congestion state: a queue threshold is used to decide whether the movement is congested or not.

The amount of cars that will be at the intersection at a time interval Dt is random. We abbreviate this number of arrivals to n and represent it as the number of arrivals through the Poisson random variable with such a rate λ . λ is the average intensity of traffic and not a constant value as the arrival rate may change with the time of year.

Arrivals are random hence the flow $q_{in}(k)$ at a given time step is also random. The output flow $q_{out}(k)$ is also uncertain since it is affected by two aspects: the existing queue length and signal choice $u(k)$ (either the phase is green or red). Then even the number of vehicles leaving the system in a step is not fixed.

Whether or not it is due to the arrivals or departures, the queue length becomes a stochastic state that varies with time.

In order to simplify the control problem, we simplify the queue to congestion states by a threshold. When the queue is above the threshold then the movement would be congested otherwise it is non- congestion.

The system is capable of transitioning among these congestion states since both arrivals as well as departures change over time. They are modeled by transition probabilities between states which are dependent on the present state and the signal action.

This develops the probabilistic organization of the model whereby uncertainty gets into the model via arrivals and moves to flow, queue, and congestion.

Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Answer: In our project, probabilistic relationship is applied to relate random traffic flow and signal control decisions.

The arrival rate n and the rate of input q_{in} does not hold any constant. They rely on the inflow rate λ and change over time to cause the uncertainty in the queue length. Owing to this, we are not exactly aware of the future situation of the system, but we can elaborate it as probabilities.

The signal phase affects the flow of output q_{out} , and the vehicles in the queue also influence the flow. Hence all control actions have a direct influence on the time step wise development of the queue.

We categorize the team lengths into a congestion level based on a threshold to come up with decisions. The greatest probabilistic relationship in the model is the transition of states. We suppose that the state of next congestion is only dependent on the present state and the current control action. This is the Markov property.

Based on this supposition, we define the transition probability.

$$P(i \rightarrow j \mid u)$$

that is, the likelihood of the system to shift in any one of the congestion states as a result of a given signal choice.

Such transition probabilities enable the controller to rank the various signal actions and select the one which causes the least probability of future congestion although they are not even sure of the exact future number of arriving vehicles.

Then not just the way the current queue reacts to the system but based on the predicted future behaviour of the traffic, the system makes decisions.

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with this project's implementation or experience. Highlight any assumptions that influenced design or evaluation choices.

Answer:

- We have characterised the traffic intersection as a stochastic system.
- We simulated the arrivals of the vehicles with a Poisson distribution and specified the way how the queue changes at each time period, depending on the arrivals and decisions of signals. The system state is the queue length and the signal phase is the control action.
- We suppose that the next state is only dependent on the present queue and present signal decision (Markov property). This enables us to describe the system as a Markov Decision Process.
- It is aimed at minimizing the anticipated congestion by choosing signal phases that regulate further increase of queues.
- At this point, the algorithm is not being implemented. We have developed the mathematical framework, which will be used in implementation and validation in the future milestones.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify its key assumptions, components, or relationships that are presented as well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your expectations.

Answer:

- During the Milestone-1, we determined the problem of congestion and conceptually explained the uncertainty of traffic flow. The emphasis was primarily on the reasons why variable traffic conditions cannot be controlled by fixed time.
- This was translated into a mathematical model in Milestone-2. To model the arrival process, we used a Poisson distribution, modeled the queue dynamics, added congestion states, and made the system a Markov Decision Process. This manifests a definite change in qualitative discourse to probabilistic modelling.
- This flow is taken into consideration in our concept map: arrivals $\rightarrow$ queue dynamics $\rightarrow$ congestion state $\rightarrow$ state transitions $\rightarrow$ control policy. It is a natural structure of previous studies, in which adaptive signal control systems have been constructed based on stochastic arrivals and Markovian transitions in the state.
- On the whole, Milestone-2 represents a more profound and systematic perception of the problem.

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Answer:

Although the mathematical structure is specified, the following probabilistic issues are not resolved:

- Poisson assumption on arrivals validation.
- variation of arrival rate λ with time.
- Markov Power of the assumption.
- Simplification of deterministic service rate.
- Choosing congestion threshold.
- Short term bursts of traffic |human|; Stability during bursts of traffic.
- Existence and structure of optimal policy.

Data validation, data simulation, and policy computation will be used to find the answers to these questions in the later milestones.

End of Submission